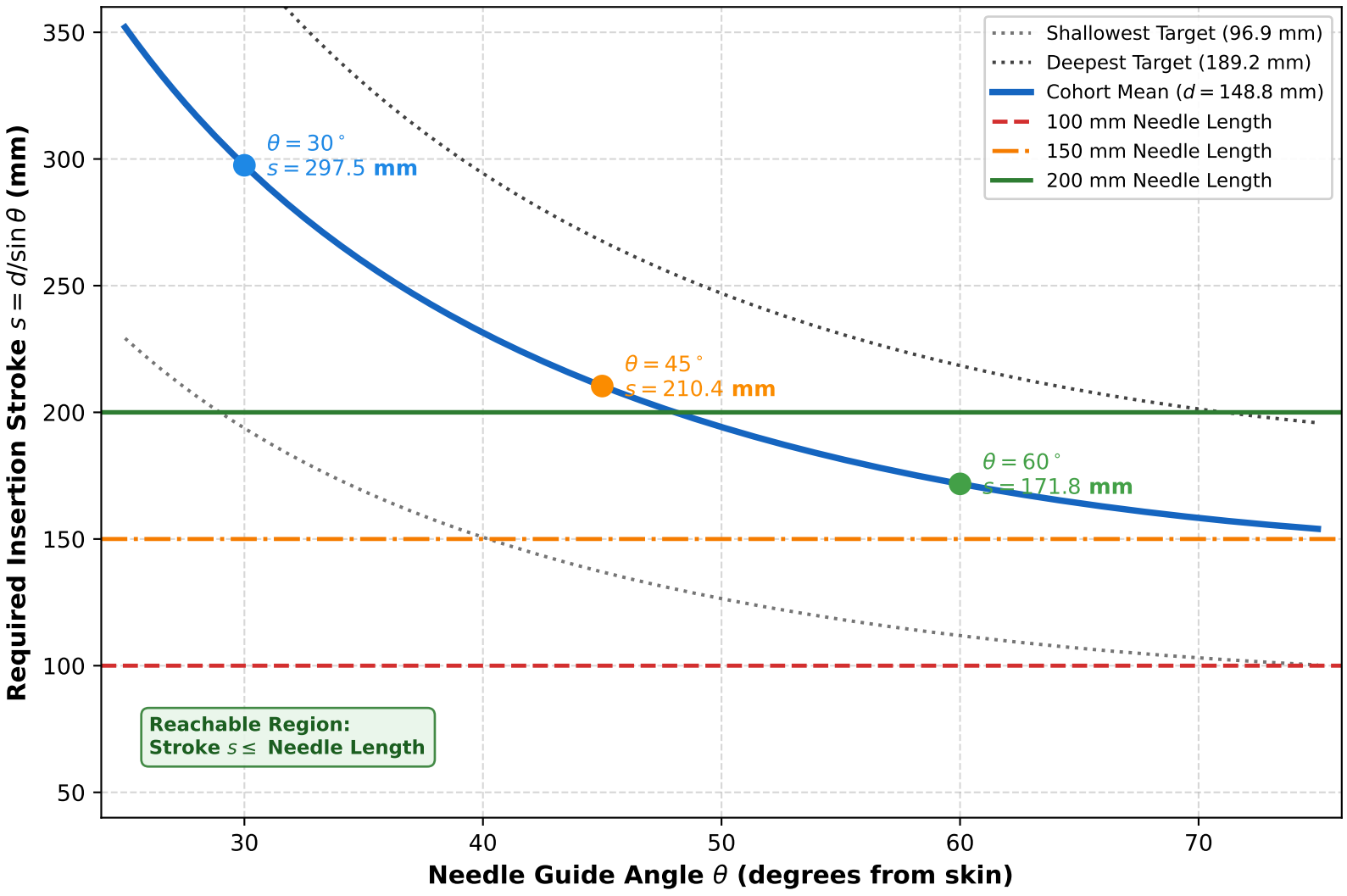
Geometric Needle Stroke vs. Needle Length Thresholds
Shallowest Target (96.9 mm)
Deepest Target (189.2 mm)
Cohort Mean ($d = 148.8$ mm)
100 mm Needle Length
150 mm Needle Length
200 mm Needle Length
$\theta = 30^\circ$
$s = 297.5$ mm
$\theta = 45^\circ$
$s = 210.4$ mm
$\theta = 60^\circ$
$s = 171.8$ mm
Reachable Region:
Stroke $s \leq$ Needle Length
Required Insertion Stroke $s = d/\sin\theta$ (mm)
Needle Guide Angle θ (degrees from skin)
350
300
250
200
150
100
50
30
40
50
60
70